

TECHNICAL REQUIREMENTS DOCUMENT (TRD)

Frontend

Framework:

- React 19
- Vite
- TypeScript

UI & Styling:

- Material UI (MUI)
- Framer Motion

State Management:

- TanStack React Query
- React Context API

Forms & Validation:

- React Hook Form
- Yup

Networking:

- Axios

Routing:

- React Router DOM

Charts:

- Recharts

Backend

Language:

- Java 21

Framework:

- Spring Boot 3

Modules:

- Spring Security
- Spring Data JPA
- Spring Validation
- Spring Mail
- Spring Actuator

Utilities:

- Lombok
- MapStruct
- JWT

Database

MySQL 8

Authentication

JWT Authentication

Access Token Expiration: 24 Hours

Password Hashing: BCrypt

Role-Based Access Control:

- ADMIN
- LEAD
- MEMBER

Hosting

Frontend: Vercel

Backend: Render

Database: Railway MySQL

File Storage

Cloudinary

Supported Uploads:

- Profile Images
- Task Attachments
- Project Documents
- PDFs
- DOCX

Third-Party Services

Email: Brevo

File Storage: Cloudinary

Monitoring: Sentry

Analytics: Recharts

Key Libraries

Frontend:

- axios
- react-query
- react-hook-form
- yup
- framer-motion
- recharts

Backend:

- spring-security
- spring-data-jpa
- spring-validation
- jjwt
- lombok
- mapstruct

Environment Variables

Frontend

VITE_API_URL

Backend

DB_URL DB_USERNAME DB_PASSWORD

JWT_SECRET JWT_EXPIRATION

MAIL_USERNAME MAIL_PASSWORD

CLOUDINARY_CLOUD_NAME CLOUDINARY_API_KEY CLOUDINARY_API_SECRET

Constraints

Must Support:

- Desktop
- Tablet
- Mobile

Browsers:

- Chrome
- Firefox
- Edge

Expected Scale:

- 5000+ users
- 100+ concurrent users

Security Requirements:

- JWT Protected APIs
- BCrypt Password Hashing
- Input Validation
- SQL Injection Prevention
- XSS Protection
- CORS Protection
- Role-Based Authorization

Performance Targets:

- API Response < 500ms
- Dashboard Load < 2 Seconds
- 99.9% Uptime